AIRSTREET · TRACEABILITY MATRIX

AirStreet traceability matrix

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DOCUMENT CONTROL

Document ID	AST-RTM-001	Version	0.3
Status	Draft	Date	2026-10-29
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/airstreet		
Source	docs/03b-traceability.md @ aa55b15		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-29	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-28	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-28	Amish Chadha	Generated by .kit/trace.py

AirStreet traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	PM2.5 at street level	AST-CAL-001, AST-STM-001, AST-OPT-001	Analysis (AST-CAL-001 E), TRL 3; CalRig chamber check and field collocation, TRL 4	At risk: within the bound up to 80 % RH; outside it at 85 % RH and above, where water uptake makes the correction uncertain
UR-002	PR-002	NO2 at street level	AST-CAL-001, AST-STM-001, AST-OPT-001	Analysis (AST-CAL-001 C), TRL 3; collocation with a chemiluminescence reference, TRL 4	At risk: about 4.3 ppb from a budget of literature and assumed terms, with the datasheet noise over 12 independent records an hour and a 5 % slope term (AST-DDR-005 F1 and the independent review); hours outside the sensor's rated range are flagged (PR-007)
UR-003	PR-003	NO2 near the WHO annual guideline	AST-CAL-001	Collocation statistics (research)	Withdrawn: annual uncertainty about 4.65 ug/m3 (1 sigma); no WHO guideline claim
UR-003	PR-004	Streets above the EU NO2 limit value	AST-CAL-001, AST-OPT-001	Analysis (AST-CAL-001 C), TRL 3; collocation statistics, TRL 4	At risk: decisive only below 33.9 or above 46.1 ug/m3 (at the directive's 20 degC). The recast directive (EU) 2024/2881 sets 20 ug/m3 from 2030 (AST-REG-001 A2), with a band of 13.9 to 26.1 ug/m3; a new target is proposed, awaiting Amish (AST-DEC-001)
UR-001, UR-002	PR-005	Temperature and humidity for corrections	AST-CAL-001, AST-BLD-001	Analysis (AST-CAL-001 D), TRL 3; CalRig and side-by-side with an aspirated reference, TRL 4	At risk: SHT45 meets the sensor figures; shield error 0.42 to 0.87 K at 1 m/s, which reads 4.1 % RH low at 85 % RH
UR-005	PR-006	Stated calibration method	AST-CAL-001	Inspection of the calibration record per node, TRL 4	Met by design (AST-DDR-001 D3)
UR-006	PR-007	Raw data for recalibration	AST-CAL-001, AST-BLD-001	Inspection of the firmware and data schema, TRL 4	Met by design: flags set when the shield air is above 35 degC, RH above 80 % (PM) or 85 % (NO2) or below 20 % (PM) or 15 % (NO2), or below 10 degC (PM)
UR-004	PR-008	Reporting for street maps	AST-CAL-001, AST-BLD-001	Analysis of timing, airtime and storage (AST-CAL-001 B), TRL 3	Met on paper: at 15 min TTN fair use is met up to SF9; 7 days in 40.3 kB
UR-013	PR-009				

User requirement	Product requirement	Title	Evidence	Verification	Status
		FieldNode power budget	AST-CAL-001, AST-BLD-001	Power budget (AST-CAL-001 A), TRL 3; energy log, TRL 4	Met on paper: 87.0 mW design load; 2.33 Wh/day against 7.75 Wh stored in the worst month; 6.6 days without sun
UR-016	PR-010	Radio duty cycle	AST-CAL-001	Airtime calculation (AST-CAL-001 B), TRL 3	Met on paper: 0.082 % at SF9, 0.60 % at SF12; 247 ms at SF9 in US915
UR-007	PR-011	Pole mounting without drilling	AST-CAL-001, AST-BLD-001	Design review and model (AST-CAL-001 F), TRL 3	Met by design: inlet 3.0 m (AST-DDR-001 D7); 140 degree V-saddles and bands cut to length
UR-009	PR-012	Survive outdoors	AST-CAL-001, AST-BLD-001, AST-STM-001	Design review (AST-CAL-001 H), TRL 3; field trial, TRL 4	At risk: the NO2 cell is rated to 40 degC and 85 % RH and the SPS30 is recommended to 40 degC and 80 % RH, so hours beyond are flagged (PR-007); condensation and insects unverified
UR-008	PR-013	Light and compact	AST-CAL-001, AST-BLD-001	Constructable model (AST-CAL-001 F), TRL 3; weighing, TRL 4	Met on paper: 3.77 kg with the sun shield (3.61 kg without), 0.23 kg margin; 0.139 m2; the pole owner's confirmation of the load is still to come
UR-011	PR-014	Levels only	AST-CAL-001	Design review, TRL 3	Met by design
UR-012	PR-015	Low cost	AST-CAL-001	Priced BOM (AST-CAL-001 I), TRL 3	Over the value-engineering target by $17.50**: sensor head $302.50; $433.50 with the FieldNode core
UR-010	PR-016	Serviceable in the field	AST-CAL-001, AST-BLD-001	Service time estimate (AST-CAL-001 H), TRL 3; timed exchange, TRL 4	Met on paper: about 14 min for a pod exchange
UR-014	PR-017	Open and garage buildable	AST-BLD-001	Inspection of the build plan AST-BLD-001 and the drawings, TRL 3	Met by design (text proposed, awaiting Amish)
UR-015	PR-018	Installation and service safety	AST-BLD-001	Inspection of the build plan safety stops and AST-RSK-001, TRL 3	Met by design (text proposed, awaiting Amish)

Gaps

None.